

FEMALE REPRODUCTIVE SYSTEM

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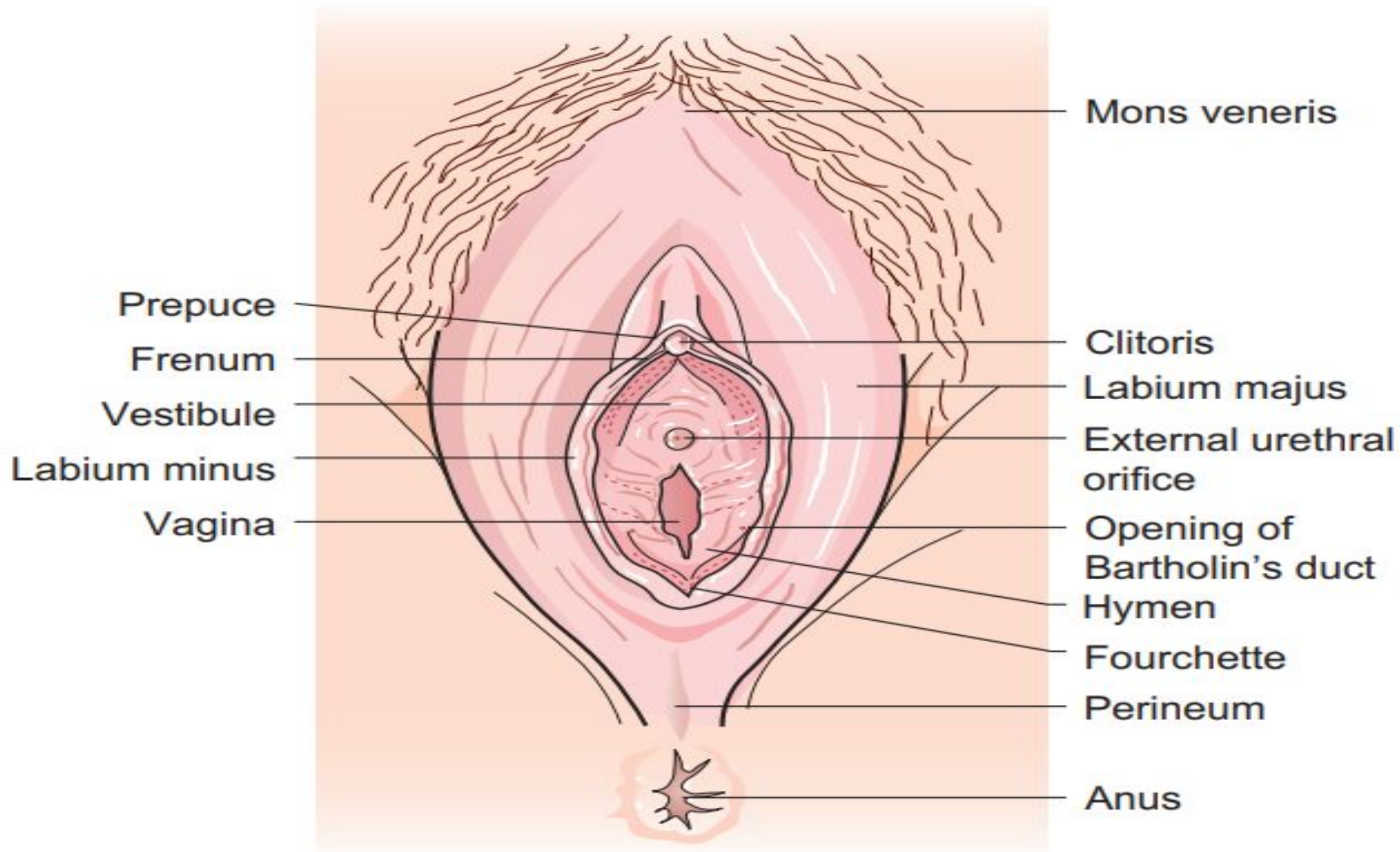
LEARNING OBJECTIVES

By the end of this lecture, students should be able to:

- ✓ Identify the major organs of female reproductive system.
- ✓ Describe the major functions of female reproductive system.
- ✓ Explain folliculogenesis including primordial follicle, primary follicle, secondary follicle & graafian follicle.
- ✓ Explain briefly the process of oogenesis.
- ✓ Describe the effects of female reproductive hormones i.e. estrogen & progesterone.

COMPONENTS OF FEMALE REPRODUCTIVE SYSTEM

- Female reproductive system consists of internal as well as external structures that supports menstruation, fertilization & pregnancy.
- The ***internal organs*** of female reproductive system consists of vagina, uterus, cervix, fallopian tubes & ovaries located within the pelvis.
- The ***external genitalia*** of female reproductive system collectively termed as ***vulva*** includes ***mons pubis, labia majora, labia minora, clitoris, vestibule & vestibular glands***(Bartholin's glands & Skene's glands).
- Vulva protects the internal organs & help with sexual reproduction.
- ***Ovaries*** are the ***primary reproductive organs*** as they produce:
 - Oocytes
 - Estrogen
 - progesterone



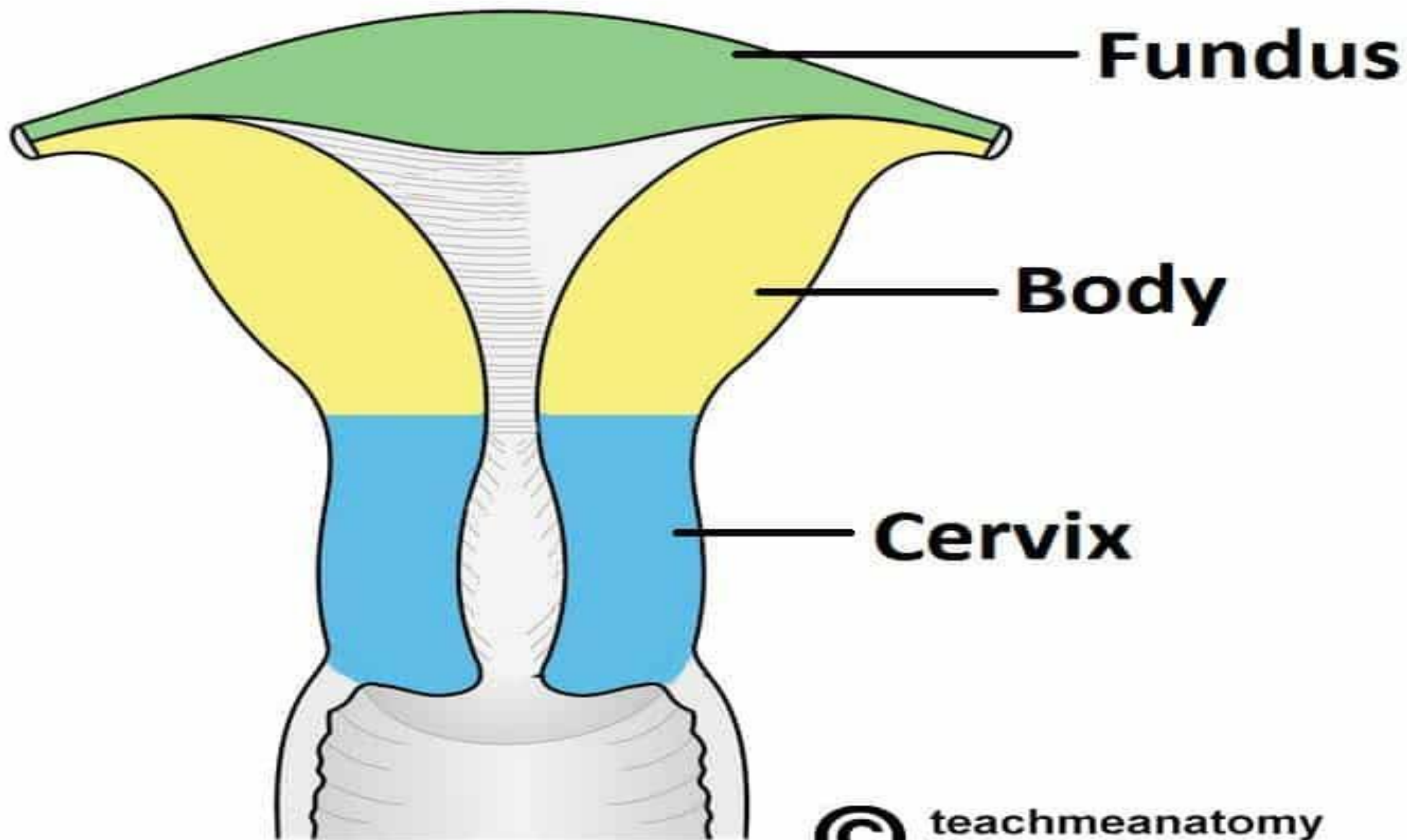
INTERNAL ORGANS OF FEMALE REPRODUCTIVE SYSTEM

VAGINA

- It is an elastic muscular canal that connects the uterus (womb) to the outside of the body.
- It acts as the channel through which the menstrual blood leaves the body during menstruation, receives penis during sexual intercourse & serves as **birth canal** allowing a baby to pass from the uterus into the world.
- It expands & self lubricates when a person is aroused.

UTERUS (WOMB)

- It is a hollow, pear shaped muscular organ about 7.5 to 8cms long where a fetus develops during pregnancy.
- It includes **fundus** (the top part), **corpus** (the main body), **isthmus** (the narrowing section) and **cervix** (the lower neck opening into the vagina).
- The wall of uterus is composed of three layers i.e. **perimetrium**, **myometrium** & **endometrium** from outside to inside.



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CERVIX

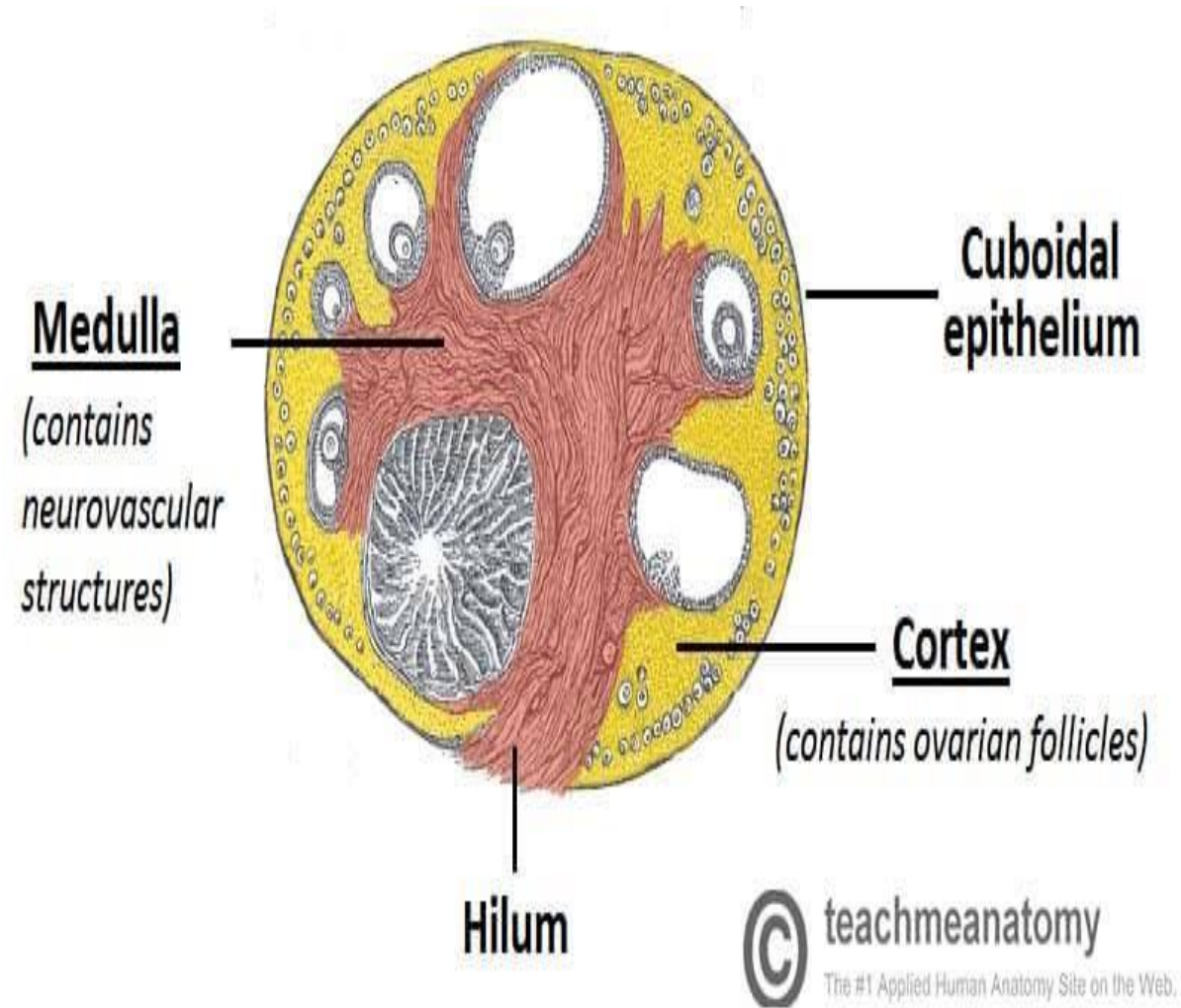
- It is the lower, narrow part of the womb roughly 2 to 3cms long that connects the uterus to the vagina.
- It is shaped like a small cylinder or a **donut** with a tiny hole in the center(**cervical os**).
- It consists of two parts, **ectocervix**(the outer part visible in the vagina) & **endocervix**(the inner canal connecting to the uterus).

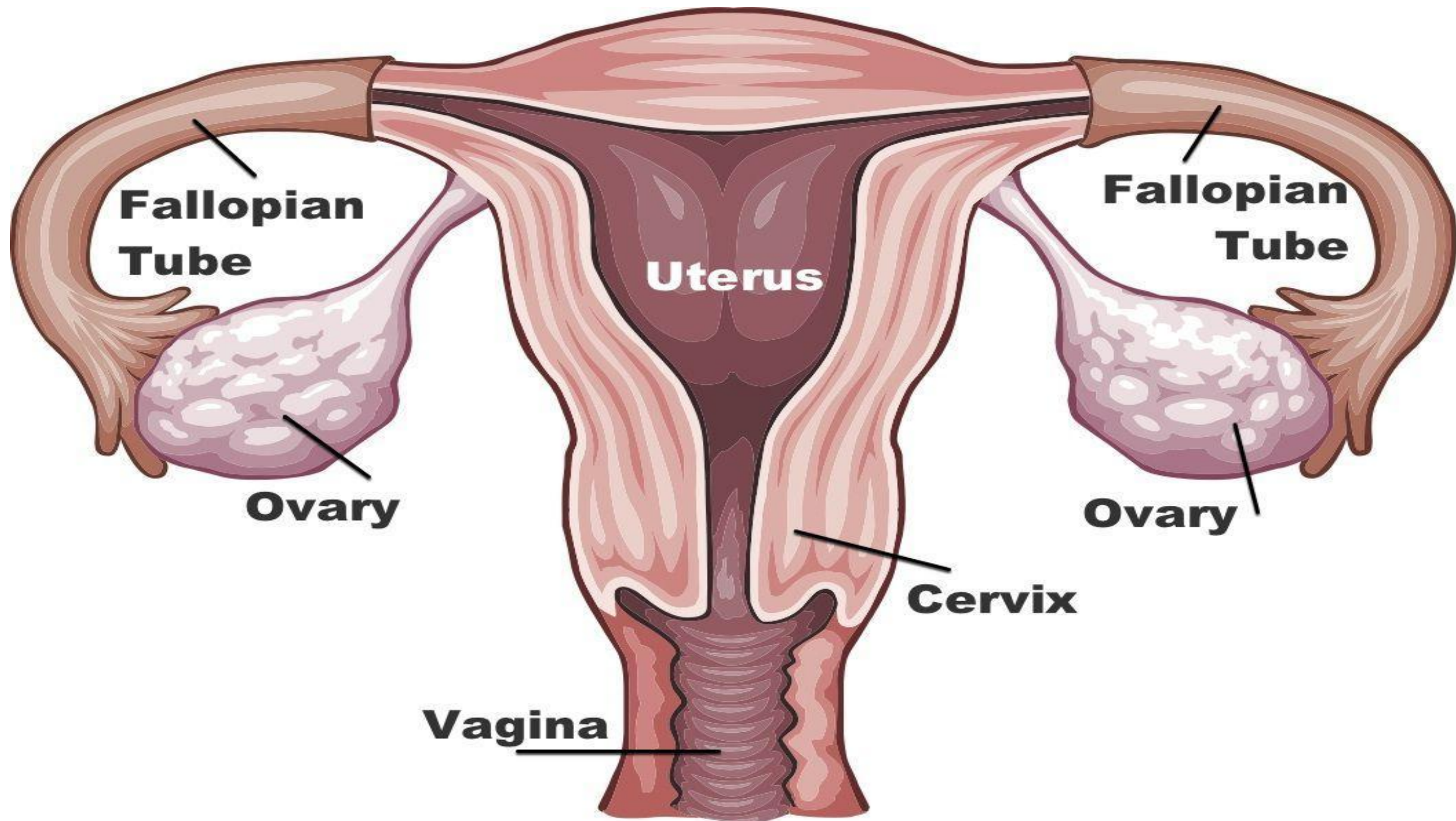
FALLOPIAN TUBES

- It is a pair of thin muscular ducts that connects ovaries to the uterus.
- They play a vital role in human reproduction by helping egg & sperm meet.
- When an ovary releases a mature egg during ovulation, finger like fringes sweep the egg inside the tube.
- The sperm swim up through the uterus into the tube where fertilization usually occurs.
- The cilia & muscle contractions then gently push the fertilized egg forward into the uterus where implantation occurs.

OVARY

- ❑ Ovaries are flattened ovoid bodies with dimensions of 4cm in length, 2cm in width & 1cm in thickness.
- ❑ Ovaries are the primary sexual organs in females & perform two functions i.e. ***gametogenic function*** (production & release of ovum) & ***endocrine function*** (secretion of female sex hormones).
- ❑ Each ovary has two portions, ***medulla***(central deeper portion of ovary) & ***cortex***(outer broader portion of ovary).





OVARIAN FOLLICLES

- Ovarian follicles are small, fluid-filled sacs inside the ovaries that contain immature eggs & produce hormones.

STAGES OF FOLLICULOGENESIS

1. Primordial follicle

- This is the earliest stage & consists of a single oocyte surrounded by a single layer of flat epithelial cells.
- Primordial follicle form before birth.

2. Primary follicle

- Flattened cells turn into columnar or cuboidal granulosa cells & a protective layer(***zona pellucida***) is formed around the egg.

Cont..

3. Secondary (antral) follicle

- Multiple layers of cells are formed & a fluid-filled space called ***antrum*** is developed.
- Thecal cells also form on the outside to produce hormones.

4. Tertiary (graafian) follicle

- These are the fully mature, large fluid-filled follicles just before ovulation.

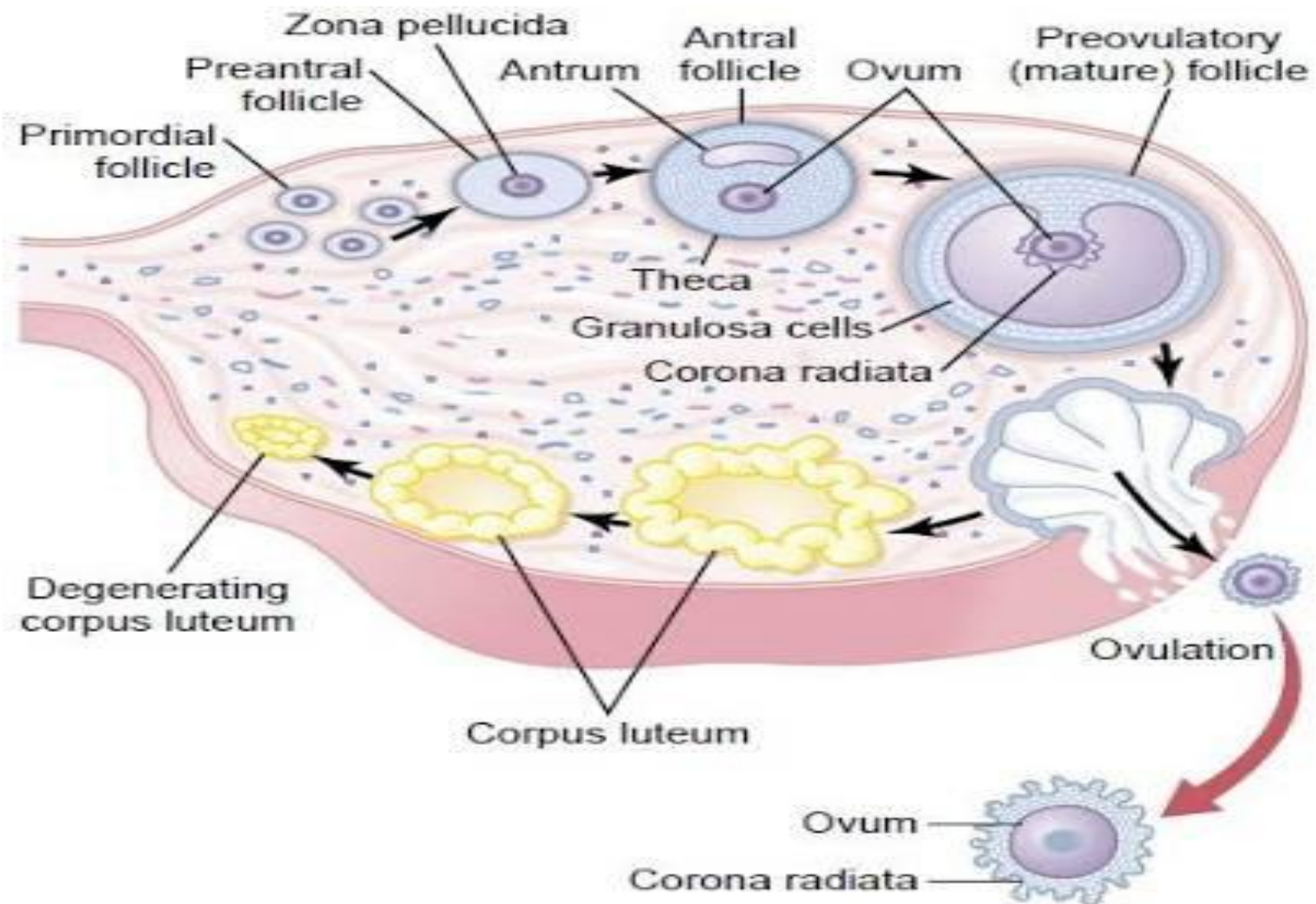


Figure 81-4

Stages of follicular growth in the ovary, also showing formation of the corpus luteum.

OOGENESIS

- It is the process of forming mature female gametes or egg cells(ova) inside the ovaries.

STAGES OF OOGENESIS

MULTIPLICATION PHASE (PRENATAL)

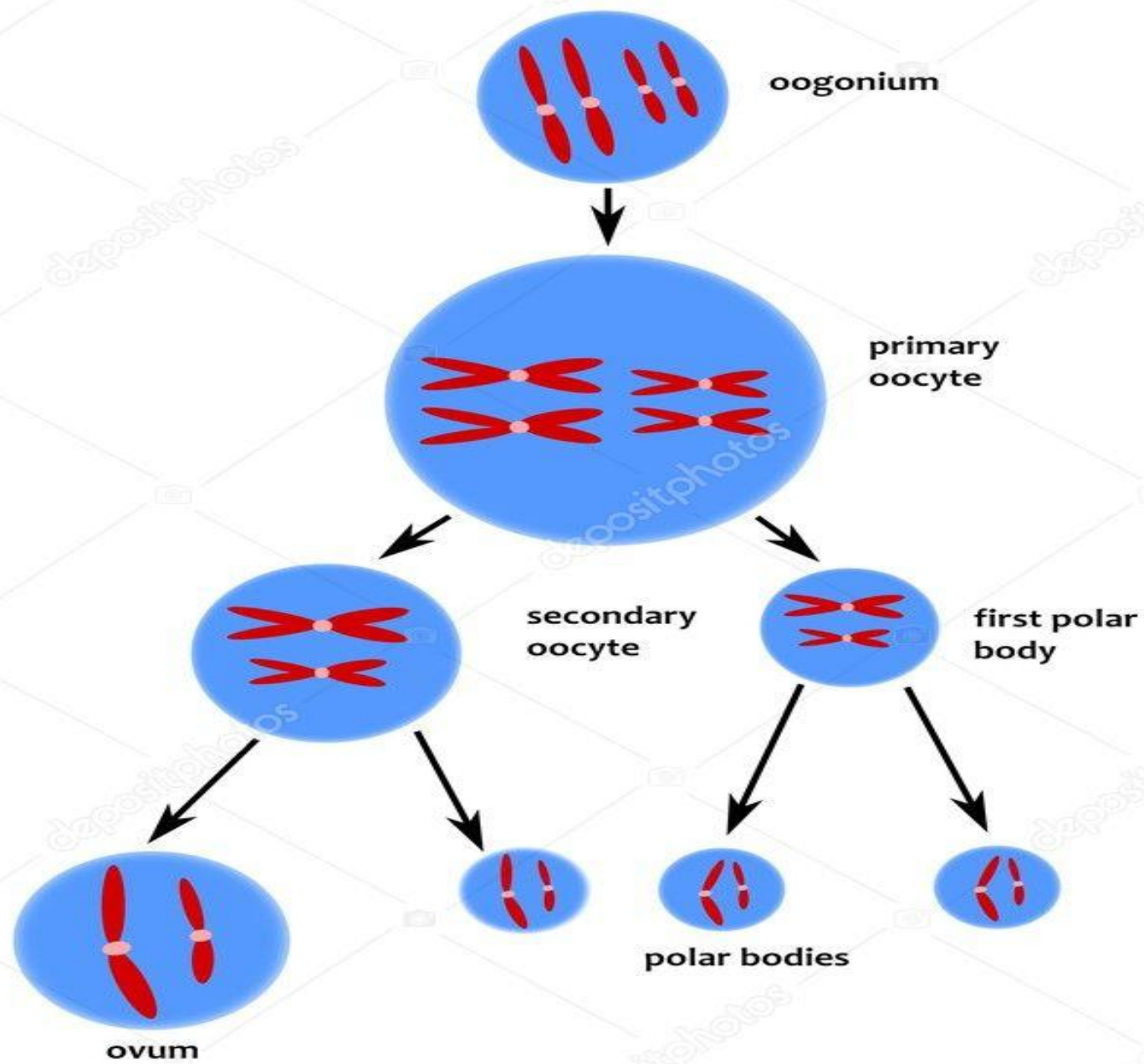
- The diploid germ cells called ***oogonia***(2n) undergo mitosis & multiply inside the fetal ovaries.
- These cells grow & enter the first phase of meiotic division to form ***primary oocytes***(2n) before birth.
- These primary oocytes then pause(arrest) in ***prophase I*** of meiotic division & remain in this form throughout the childhood.

MATURATION PHASE (PUBERTY TO MENOPAUSE)

- Hormones(LH) trigger a primary oocyte to resume meiosis I during each menstrual cycle.
- This division is uneven(asymmetrical cytokinesis) & forms one large ***secondary oocyte***(n) & one tiny ***first polar body***(n).
- The secondary oocytes enter meiosis II but pauses again at this stage.
- This secondary oocyte is released from the ovary during ovulation.

COMPLETION PHASE (UPON FERTILIZATION)

- Meiosis II completes only when a sperm penetrates the secondary oocyte & forms ***ovum***(n) & a ***second polar body***(n).
- The nuclei of sperm & ovum fuse to form a diploid ***zygote***(2n).



OVARIAN HORMONES

- Ovaries secrete female sexual hormones ***estrogen*** & ***progesterone***.
- Ovaries also secrete few more hormones like ***inhibin***, ***relaxin*** & small quantities of ***androgens***.

ESTROGEN

- In a normal non-pregnant woman, estrogen is secreted in large quantity by ***theca interna cells*** of ovarian follicles & in small quantity by ***corpus luteum*** of the ovaries.
- A small quantity of estrogen is also secreted by adrenal cortex & in pregnant woman a large amount of it is secreted by the placenta.
- Estrogen is present in three forms in plasma namely ***beta-estradiol***, ***estrone*** & ***estriol***.
- Two types of estrogen receptors are present on target cells, ***alpha estrogen receptors***(present in uterus, liver, heart & kidneys) & ***beta estrogen receptors***(present in ovaries & other tissues).
- Estrogen is transported mainly by the plasma protein ***albumin*** whereas, small quantity is also transported by ***globulin***.

FUNCTIONS OF ESTROGEN

- Major function of estrogen is to promote cellular proliferation & tissue growth in the sexual organs & in other tissues related to reproduction.
- In childhood, estrogen is secreted in small quantity while during puberty, its secretion increases significantly resulting in changes in sexual organs.

Its main functions are:

EFFECT ON OVARIAN FOLLICLES

- Promotes the growth of ovarian follicles by increasing proliferation of follicular cells.
- It also increases the secretory activity of theca cells.

EFFECT ON UTERUS

- Enlargement of uterus to about double of its childhood size due to proliferation of endometrial cells.
- Increase in the blood supply of endometrium as a result of dilatation & proliferation of endometrial blood vessels.
- Glycogen & fats deposition in endometrium.

Cont..

- Proliferation, dilatation & increased blood flow to the endometrial glands.
- Increase in spontaneous activity of uterine muscles & their sensitivity to oxytocin.
- Increase in the contractility of uterine muscles.

EFFECT ON FALLOPIAN TUBES

- Increases the no. & size of epithelial cells particularly the ***ciliated epithelial cells*** lining the fallopian tubes.
- Increase in the ciliary activity to facilitate the movement of ovum in the fallopian tube.
- Enhanced proliferation of glandular tissues in the fallopian tubes.

EFFECT ON VAGINA

- Increase in proliferation of vaginal epithelium & change in vaginal epithelium from cuboidal to stratified type as the stratified epithelium is more resistant to trauma & infection.
- Reduction in vaginal pH making it more acidic.

EFFECT ON BREAST

- Growth & deposition of fat in the extensive ductile system of breasts as well as development of stromal tissues.
- Estrogen causes development of lobules & alveoli of breasts to some extent, however progesterone is necessary for full growth of breasts & ***prolactin*** is necessary for breast function.

EFFECT ON BONES

- Estrogen increases osteoblastic activity of bones resulting in enormous increase in growth rate at the time of puberty.
- It also causes early fusion of epiphysis with shaft resulting in early cessation of growth in females than in males.
- In old age, estrogen secretion stops or diminishes significantly resulting in ***osteoporosis*** of bones making them susceptible to fractures.

EFFECT ON METABOLISM

- Increases total body protein by inducing protein anabolism.
- Causes deposition of fat in the subcutaneous tissues, breasts, buttocks & thighs.

EFFECT ON ELECTROLYTE BALANCE

- Estrogen causes sodium & water retention from the renal tubules particularly during pregnancy.

EFFECT ON SECONDARY SEXUAL CHARACTERS

- Causes hair growth in axilla & pubic region & also scalp hair grows profusely.
- Skin becomes soft & smooth & increase in vascularity of the skin.
- Affect body shape as shoulders become narrow, hip broadens, thighs converge & arms diverge.
- Fat deposition increases in breasts & buttocks.
- Pelvis becomes broad with round or oval shape in contrast to males in which pelvis is funnel shaped.

PROGESTERONE

- In non-pregnant woman, small quantity of progesterone is secreted by theca interna cells of ovaries during follicular phase but a large quantity of progesterone is secreted during secretory phase by the corpus luteum.
- In pregnant woman, large quantity of progesterone is secreted by the corpus luteum in the 1st trimester whereas, corpus luteum degenerates in the 2nd trimester & placenta secretes large amount of progesterone in 2nd & 3rd trimester.
- Progesterone is transported in the blood by plasma proteins i.e. ***albumin & globulin.***

FUNCTIONS OF PROGESTERONE

- Progesterone is concerned mainly with the final preparation of the uterus for pregnancy & the breasts for lactation.

EFFECT ON FALLOPIAN TUBES

- It promotes the secretory activities of muscular lining of the fallopian tubes.
- Fallopian tube secretions are necessary for nutrition of fertilized ovum before implantation.

EFFECT ON UTERUS

- Progesterone promotes the secretory activities of uterine endometrium during the secretory phase of menstrual cycle so the uterus is prepared for implantation of ovum.
- It causes increase in thickness of endometrium, increased blood supply to the endometrium, increase in size & secretory activities of uterine glands.
- It also causes decrease in frequency of uterine contractions during pregnancy.

EFFECT ON CERVIX

- Increases the thickness of cervical mucosa & thus inhibits the transport of sperm into the uterus.
- This effect is utilized in the contraceptive action of mini pills.

EFFECT ON MAMMARY GLANDS

- Promotes the development of lobules & alveoli of mammary glands & causes the breasts to swell due to increased secretory activity & fluid accumulation in the subcutaneous tissue.

EFFECT ON HYPOTHALAMUS

- Inhibits the release of FSH & LH through negative feedback effect.

THERMOGENIC & RESPIRATORY EFFECT

- It increase body temperature & ventilation after ovulation & during pregnancy.

EFFECT ON ELECTROLYTE BALANCE

- It increases reabsorption of sodium & water from renal tubules however in large doses cause excretion of sodium & water as it binds with the same receptors as for aldosterone.

HYPOTHALAMIC-PITUITARY-OVARIAN (HPO) AXIS

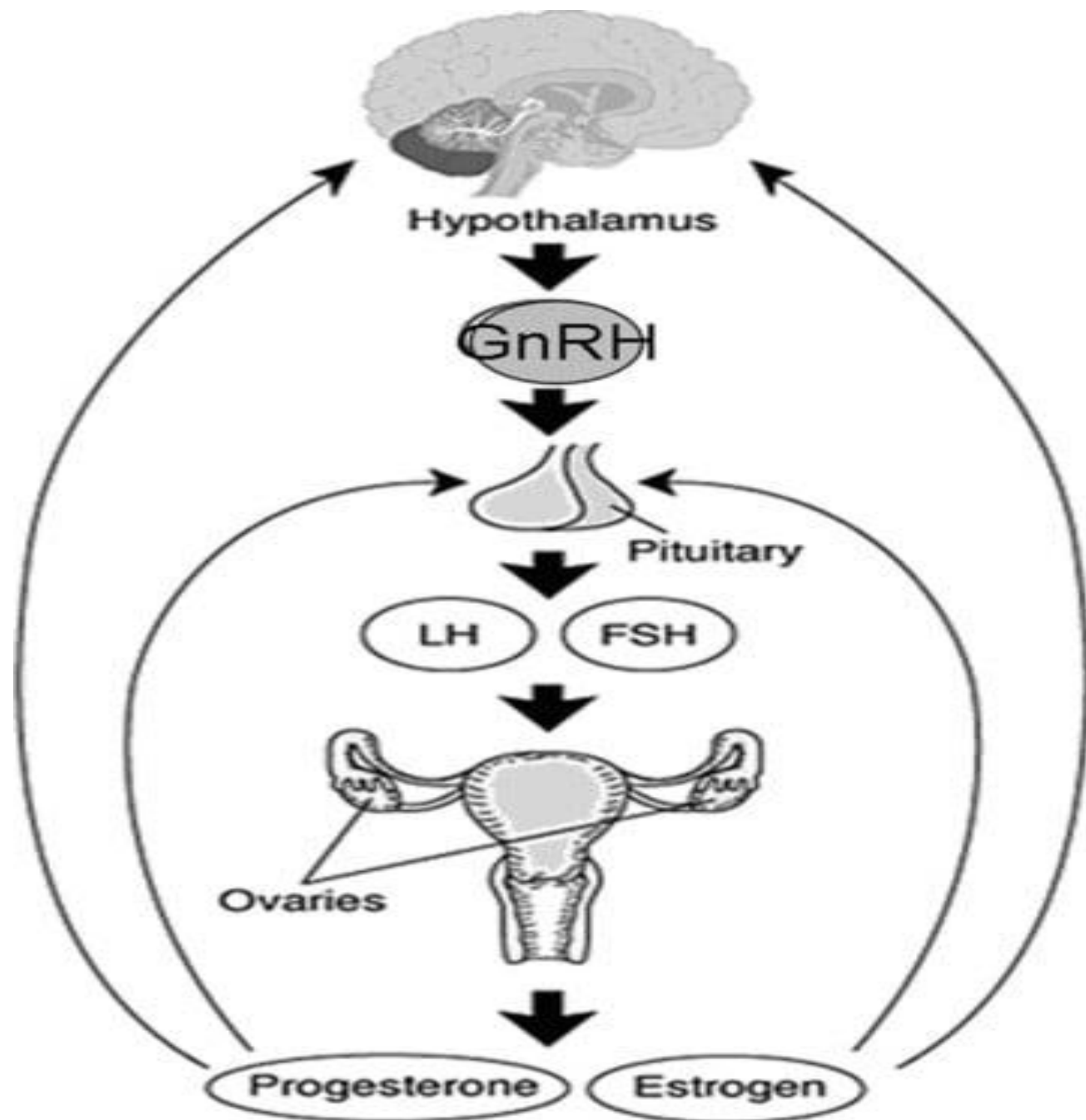
- It is a hormonal feedback loop that controls female reproduction, menstrual cycle & ovulation.

NEGATIVE FEEDBACK

- Hormones produced by the ovaries i.e. estrogen & progesterone signal back to the brain.
- Normal levels of estrogen & progesterone signal the hypothalamus & pituitary gland to slow down hormone release(*negative feedback*).

POSITIVE FEEDBACK

- Right before ovulation, high levels of estrogen exerts opposite effect & trigger a high surge in LH, & FSH(*positive feedback*) which causes release of eggs.



MAJOR FUNCTIONS OF FEMALE REPRODUCTIVE SYSTEM

- The main function of female reproductive system is to produce egg cells, enable fertilization, support the developing fetus & produce sex hormones.

PRODUCTION OF FEMALE GAMETES

- The process of formation of female gametes is called oogenesis, where diploid cells develop into mature haploid eggs through meiosis.

PRODUCTION OF REPRODUCTIVE HORMONES

- Female reproductive hormones are mainly produced by the ovaries, the pituitary gland & the hypothalamus in a highly regulated feedback loop.
- ***Estrogen & progesterone*** are mainly produced in the ovaries(& in placenta during pregnancy) to drive menstrual cycle & prepare the body for pregnancy.
- ***Follicle stimulating hormone(FSH) & luteinizing hormone(LH)*** are produced & released by pituitary gland at the base of the brain to stimulate egg maturation & trigger ovulation.

Cont..

- **Gonadotropin-releasing hormone(GnRH)** is produced in the hypothalamus & signals the pituitary gland to release FSH & LH.
- **Androgens** & small amounts of **testosterone** are produced by the ovaries & the adrenal glands.

RECEPTION OF SPERM & FERTILIZATION

- Vagina serves as the initial reception point whereas cervix functions as a selective barrier allowing only the healthiest sperms to pass into the uterus.
- Muscular contractions within the uterus propel the sperm towards the fallopian tubes(actual site of fertilization).

TRANSPORT OF GAMETES & EMBRYO

- Tiny hair-like projections(cilia) & rhythmic muscular contractions gently move the egg or resulting embryo through the fallopian tube towards the uterus.

IMPLANTATION

- It is the early stage of pregnancy in which the fertilized egg(embryo) attaches to the inner lining of uterus(endometrium).

MAINTENANCE OF PREGNANCY

- Uterus provides environment for fetal development.

PARTURITION

- It is the process of delivering the fetus from mother's uterus at the end of pregnancy.
- Uterine contractions participates in childbirth.

LACTATION

- Mammary glands produce milk after childbirth.

Structure	Function
Ovaries	Produce the hormone estrogen and progesterone Production of eggs
Fallopian tubes (oviducts)	Carry the ovum from the ovary to the uterus Usually the site of fertilization
Fimbria	Sweep the ovum into the oviduct following ovulation
Uterus (womb)	Pear-shaped organ in which the embryo and fetus develop. Involved in menstruation
Cervix	Separates the vagina from the uterus Holds the fetus in place during pregnancy Dilates during birth to allow the fetus to leave the uterus
Vagina	· Extends from the cervix to the external environment · Provides a passageway for the sperm and menstruation flow · Functions as the birth canal.

POST ASSESSMENT

Q1. Which structure develops from the ruptured ovarian follicle after ovulation?

- a) Corpus albicans
- b) Corpus luteum
- c) Graafian follicle
- d) Primary follicle
- e) Theca cells

Q2. Which layer of uterus undergo major cyclic changes during the menstrual cycle?

- a) Perimetrium
- b) Myometrium
- c) Endometrium
- d) Parametrium
- e) Serosa

Cont..

Q3. Which hormone is mainly responsible for the development of female secondary sexual characteristics?

- a) FSH
- b) LH
- c) hCG
- d) Estrogen
- e) Progesterone

Q4. During oogenesis, primary oocytes are formed during:

- a) Puberty
- b) Menopause
- c) Fetal life
- d) Fertilization
- e) Pregnancy